

第一章

P25 T2. (1) A发生, B与C不发生 $A\bar{B}\bar{C}$

(2) A与B都发生, 而C不发生 $AB\bar{C}$

(3) A, B, C中至少有一个发生

$A \cup B \cup C$

(4) A, B, C都发生 ABC

根据德摩根律

"或" $\rightarrow$ "与"

(5) A, B, C都不发生 $\bar{A}\bar{B}\bar{C}$

"或" $\rightarrow$ "与"

(6) A, B, C中不多于一个发生 $= \bar{A}\bar{B}\bar{C} \cup A\bar{B}\bar{C} \cup \bar{A}B\bar{C} \cup \bar{A}\bar{B}C$

(7) A, B, C中不多于两个发生 (0, 1, 2) 除去三个都发生的情况

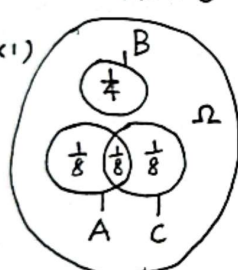
$A \cup B \cup C - ABC = \bar{A}\bar{B}\bar{C} \cup A\bar{B}\bar{C} \cup \bar{A}B\bar{C} \cup \bar{A}\bar{B}C$

这是1个或2个发生

(8) A, B, C中至少有两个发生

$AB \cup AC \cup BC$

3. (1)



A, B, C至少有一个发生事件为

$A \cup B \cup C$, 概率为

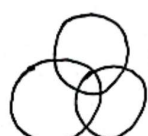
$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(AB) - P(BC) - P(AC) + P(ABC) \\ &= \frac{1}{4} + \frac{1}{4} + \frac{1}{4} - \frac{1}{8} - \frac{1}{8} - \frac{1}{8} + \frac{1}{8} = \frac{5}{8} \end{aligned}$$

(2) $A \cup B$ 的概率

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(AB) \\ &= \frac{1}{2} + \frac{1}{3} - \frac{1}{10} = \frac{5}{6} - \frac{1}{10} = \frac{50}{60} - \frac{6}{60} = \frac{44}{60} = \frac{11}{15} \end{aligned}$$

(3) $\bar{A}\bar{B}$ 的概率

$$\begin{aligned} P(\bar{A}\bar{B}) &= 1 - P(A \cup B) \\ &= \frac{4}{15} \end{aligned}$$



(4) $A \cup B \cup C$ 的概率

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(AB) - P(BC) - P(AC) + P(ABC) \\ &= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} - \frac{1}{6} - \frac{1}{6} - \frac{1}{6} + \frac{1}{6} = \frac{1}{6} \end{aligned}$$

$$P(\bar{A}\bar{B}\bar{C}) = P(\Omega) - P(A \cup B \cup C) = 1 - \frac{17}{20} = \frac{3}{20}$$

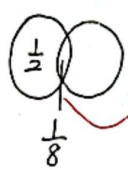
$$\begin{aligned} P(\bar{A}\bar{B}C) &= P(C) - P(AC) - P(BC) + P(ABC) \\ &= \frac{1}{5} - \frac{1}{15} - \frac{1}{15} + \frac{1}{30} \\ &= \frac{2}{30} - \frac{1}{30} = \frac{1}{30} \end{aligned}$$

$$\begin{aligned} P(\bar{A}B\bar{C}) &= P(\bar{A}B) + P(AC) + P(BC) - P(ABC) \\ &= \frac{1}{5} + \frac{1}{15} + \frac{1}{15} - \frac{1}{30} \\ &= \frac{1}{3} + \frac{1}{30} - \frac{1}{30} = \frac{2}{30} = \frac{1}{15} \end{aligned}$$

(3) $P(A) = \frac{1}{2}$ (i) A, B互不相容

$P(\bar{A}\bar{B}) = P(A) = \frac{1}{2}$

(ii) $P(AB) = \frac{1}{8}$ 求 $P(\bar{A}\bar{B})$



$$\begin{aligned} P(\bar{A}\bar{B}) &= P(A) - P(AB) \\ &= \frac{3}{8} \end{aligned}$$



5. 10片药片中有5片是安慰剂

(1) 所求概率为 $p = 1 - P(\text{取到的5片都不是安慰剂}) = P(\text{取到一片安慰剂})$

$$= 1 - \frac{1}{C_5^{10}} = \frac{C_4^9}{C_5^{10}} = \frac{126}{126}$$

(2) 从中每次取一片, 作不放回抽样, 求前3次都取到安慰剂的概率

$$p = \frac{5}{10} \times \frac{4}{9} \times \frac{3}{8} = \frac{1}{12}$$

14. (1) $P(\bar{A}) = 0.3$ $P(B) = 0.4$ $P(A\bar{B}) = 0.5$

$$P(B|A\cup\bar{B}) = \frac{P(B \cap (A\cup\bar{B}))}{P(A\cup\bar{B})} = \frac{P(AB)}{P(A\cup\bar{B})} = \frac{1}{4}$$

$$P(A\cup\bar{B}) = P(A) + P(\bar{B}) - P(A\bar{B}) = 1 - P(\bar{A}) + 1 - P(B) - P(A\bar{B}) \\ = 1 - 0.3 + 1 - 0.4 - 0.5 = 0.8$$

$$P(AB) = P(A) - P(A\bar{B}) = 0.2$$

(2) $P(A) = \frac{1}{4}$ $P(B|A) = \frac{1}{3}$ $P(A|B) = \frac{1}{2}$

$$P(A\cup B) = P(A) + P(B) - P(AB) \\ = P(A) + \frac{1}{4} + \frac{1}{6} - \frac{1}{12} = \frac{1}{3}$$

$$P(A|B) = \frac{P(AB)}{P(B)} \quad P(B|A) = \frac{P(AB)}{P(A)} \quad P(AB) = P(A)P(B|A) \\ = \frac{1}{4} \times \frac{1}{3} = \frac{1}{12}$$

$$P(B) = \frac{P(AB)}{P(B|A)} = \frac{\frac{1}{12}}{\frac{1}{2}} = \frac{1}{6}$$

16. 记孩子得病为事件A. 母亲得病|孩子得病 = $B|A$

父亲得病为事件C. 父亲得病|母亲及孩子得病 = $C|AB$

母亲及孩子得病但父亲未得病 $AB\bar{C}$

$$P(A) = 0.6 \quad P(B|A) = 0.5 \quad P(C|AB) = 0.4$$

$$P(AB) = P(B|A)P(A) = 0.3 \quad P(C|AB)P(AB) = P(ABC) = 0.3 \times 0.4 = 0.12$$

$$P(AB\bar{C}) = P(AB) - P(ABC) = 0.3 - 0.12 = 0.18$$

还有1分

18. 拨号不超过三次而接通所需电话的概率 $p = \frac{1}{9} + \frac{8}{9} \times \frac{1}{8} + \frac{8}{9} \times \frac{7}{8} \times \frac{1}{7} = \frac{1}{3}$

已知最后一个数字是奇数 $p = \frac{1}{5} + \frac{4}{5} \times \frac{1}{4} + \frac{4}{5} \times \frac{3}{4} \times \frac{1}{3} = \frac{3}{5}$

$$p = \frac{1}{10} + \frac{9}{10} \times \frac{1}{9} + \frac{9}{10} \times \frac{8}{9} \times \frac{1}{8} \\ = \frac{3}{10}$$

21. 是色盲患者是为事件A $P(A) = 5\% \times \frac{1}{2} + 0.25\% \times \frac{1}{2}$

是男性为事件B. 是色盲患者且是男性

$$P(B|A) = \frac{5\% \times \frac{1}{2}}{5\% \times \frac{1}{2} + 0.25\% \times \frac{1}{2}} \\ = \frac{5}{5.25} = \frac{1}{1.05} = \frac{100}{105} = \frac{20}{21}$$



24. (1) 第一次取到零件是一等品为事件A 第一次箱为事件B 取第二箱为事件C

$$P(A) = \frac{1}{2} \times \frac{10}{50} + \frac{1}{2} \times \frac{18}{50} = \frac{1}{10} + \frac{9}{50} = \frac{1}{10} + \frac{3}{10} = \frac{2}{5}$$

(2) 两次取到一等品为事件AD 在第一次取到零件是一等品条件下, 第二次取到也是一等品

$$P(AD|A) = \frac{P(AD)}{P(A)} = \frac{\frac{1}{2} \times \frac{10}{50} \times \frac{9}{49} + \frac{1}{2} \times \frac{18}{50} \times \frac{17}{49}}{\frac{2}{5}} = \frac{\frac{1}{2} \times \frac{9}{49} + \frac{51}{49}}{\frac{2}{5}} = \frac{\frac{69}{98}}{\frac{2}{5}} = \frac{69}{162} = 0.6856$$

26. (1) 主人回来时树还活着为事件A 浇水为事件B 不浇水为事件 $\bar{B}$ 则树活着 $P(A|B) = 1 - P(\bar{A}|B) = 0.2$

$$0.1 \times 0.2 + 0.9 \times 0.85$$

$$P(A|\bar{B}) = 1 - P(\bar{A}|\bar{B}) = 1 - 0.15 = 0.85$$

$$P(B) = 0.1 \quad P(\bar{B}) = 0.9$$

$$P(A) = P(B)P(A|B) + P(\bar{B})P(A|\bar{B}) = 0.1 \times 0.2 + 0.9 \times 0.85 = 0.02 + 0.765 = 0.785$$

(2) 若主人回来时树已死去, 邻居忘记浇水可表示为 $P(\bar{B}|\bar{A})$, $P(\bar{A}) = P(B)P(\bar{A}|B) + P(\bar{B})P(\bar{A}|\bar{B}) = 1 - P(A)$

$$P(\bar{B}|\bar{A}) = \frac{P(\bar{B}\bar{A})}{P(\bar{A})} = \frac{0.1 \times 0.8}{0.1 \times 0.8 + 0.9 \times 0.15} = \frac{0.08}{0.215} = \frac{8}{215} = 0.372$$

